Quarterly Health Bulletin, automated

Recruitment assignment, Forward Deployed Engineer

Ministry of Health, Rwanda, scenario

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Executive summary

The choice. Problem A, the Quarterly Health Bulletin, automated end to end, with figure-level lineage built in from the start rather than added later. B needs facility-level freshness that monthly aggregate reporting on a two to three week lag cannot supply, which is a cadence gap and a separate project. C matches TB and HIV records by probability, since the systems share no patient identifier, so a wrong match puts one patient's HIV status into another's treatment decision, and no MoH process exists to review and sign that risk off inside six weeks. I would propose a weekly reconciliation report instead of record merging. A is the reachable wedge, not the most valuable outcome available, and Deliverable 1 says so in those words.

What is built. A working pipeline that reads the five provided CSVs, resolves every column through a declared crosswalk, and publishes all four 2024 quarters as self-contained HTML with no client-side JavaScript. Python with Hamilton, DuckDB and Parquet for the data layer; Astro with Flint compiled to Vega-Lite for the render. Five automated checks gate publication; four of them exist because they caught a real defect, and the fifth is preventive and labelled as such rather than counted as a catch. Setup and run instructions are in the accompanying repository.

All four editions are live at **https://sand-fde-bulletin.pages.dev**. The pages carry no client-side JavaScript and no external assets, so they render identically offline from the repository. Four representative pages are reproduced at the end of this document; the site is the whole thing.

What the data actually says, which is not what it looks like. Two findings shaped the bulletin more than any metric in the brief did.

The largest single category of neonatal death in this dataset is not a clinical cause. "Other" accounts for 30.7% of resolved deaths across the year, and it is the largest bucket in every one of the ten months held, ranging from 28.8% to 32.1%. The two named causes behind it cannot be separated: birth asphyxia takes 23.7% and prematurity 23.6%, a gap of 15 deaths in 9,036, and which of the two leads flips between months. So a bulletin that ranked named clinical causes would be reporting noise, and one that led with asphyxia as the top killer would be wrong twice over. The finding is about the reporting system rather than about newborn medicine, and it is the more actionable of the two.

A composite equipment index correlates about -0.87 with facility mortality across all 117 facilities, which reads as "equip facilities, save newborns." Stratified by facility tier it collapses to near zero. The pooled number describes which tier a facility sits in, not what equipping it would do, and the within-tier residual is too small and too unstable across index compositions to support a claim. A bulletin that had published the pooled figure would have pointed a Ministry's budget at a confound. Every covariate in the bulletin is therefore shown pooled and within tier, with the caveat attached rather than the section withheld.

What it does not do. It does not correct source data, redefine any indicator, touch patient-level data, or attempt real-time anything. The 2024-01 and 2024-03 batch conflict is disclosed, not resolved, because nothing in the file establishes which load is authoritative and inventing a rule would be worse than naming the problem.

One thing to hold in mind throughout. The provided data is synthetic. Facility mortality of 14 to 73 per 1,000 live births runs three to four times real Rwandan rates, February and December are absent, and the facility coordinates are uniform random inside the country's bounding box. The pipeline, the checks and the reasoning are the deliverable. The health findings are not claims about Rwanda.

Deliverable 1, Problem Decomposition and Scoping

Role: Forward Deployed Engineer · **Country:** Rwanda · **Sprint:** 6 weeks, 2 FDEs **Sponsor:** MoH Director · **Internal:** Solutions Manager, Country Director, Operations Specialist

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How to read this

Section 1 is what I would do in Week 1 before proposing anything. Section 2 is the commitment I would make **at the Week 1 gate**, conditional on the gate passing. Everything is written to be falsifiable.

Assumptions are flagged **[A#]** and collected in 2.5.

[A1] The country is Rwanda, the brief says "one of our expansion countries (i.e. Rwanda)" and the provided data uses Rwandan facility names, districts and RWF. **[A2]** I have no access to a live Sand or MoH environment. Everything below is built against the provided data and public sources, and I have tried not to convert either into an existence proof.

Section 1, Discovery Process

1.1 "Our data is a mess" is a symptom report, not a problem statement

The sentence contains at least four distinct failure classes, implying four different projects:

FAILURE CLASS	WHAT IT MEANS	WHAT IT WOULD IMPLY
Absence	Never captured at all	Digitisation problem
Latency	Captured, arrives too late to act on	Pipeline problem. DHIS2 runs 2–3 weeks behind
Distrust	Arrives on time, nobody believes it	Provenance problem. Sources disagree, nobody reconciles
Last mile	Believed, never reaches a decision	Workflow problem

Week 1's job is to find which dominates. **I do not know which it is, and the phrase does not tell me.** "We cannot make good decisions" is compatible with all four, it could mean missing data, late data, inaccessible data, untrusted data, weak analysis, or absent decision authority. Distrust is the hypothesis I would test *first* because it is the cheapest to check and the most commonly missed, not because the sentence establishes it.

1.2 Who I would talk to, and in what order

DAY	WHO	THE QUESTION THAT ACTUALLY MATTERS
1	Solutions Manager, Country Director	*Why were the use cases chosen before discovery?* What does the MoU commit us to in writing, and what has been promised to whom?
1–2	**MoH Director** (sponsor)	"*Tell me about a decision last quarter you'd have made differently with better data.*" Then: "*When the numbers you're shown disagree with what you believe, which do you act on?*"
2	**The bulletin analyst**	The most important interview. Not "what do you need", "*show me last quarter's, and open the files you built it from.*"
2–3	DHIS2 / HMIS focal point	*Where does the 2–3 week delay actually accrue?* And critically: **how do the paper facilities report today?** (see 1.3)
3	2 district health officers, one strong district, one weak	"*What did you look at immediately before your last resource decision?*" If the answer isn't data, B solves the wrong problem
3–4	1 HealthTrack hospital, 1 OpenMRS clinic, 1 paper-only facility, the **data clerk**, not the medical director	Watch month-end reporting happen
4	TB and HIV programme managers	Size C honestly: how many co-infected, how identified today, what happens when one is missed
4	**MoH ICT / InfoSec**	Hosting, data residency, service accounts, approval lead times. See 2.4, this is the classic six-week killer
5	Other FDE, Central FDEs, Product	What patterns already exist. Do not rebuild

Question discipline. Every question is **counterfactual** ("what did you do last month") or **observational** ("show me"), never **solicitational** ("what would you like"). Solicitation returns a feature list reflecting what the person thinks you can build.

One question for everyone, because the answers will not match:

> "*When two systems give different numbers for the same thing, what happens?*"

1.3 The reporting-status question I got wrong, and would now ask first

My first draft treated "paper-only facility" as equivalent to "absent from DHIS2." **That is probably false, and it was load-bearing.** A facility can keep paper clinical registers and still submit a monthly aggregate HMIS form that a district data clerk keys into DHIS2. That is the standard model across most African HMIS deployments.

So Week 1 must establish the actual distribution across at least six states, not one:

1. No patient-level digital system, **but aggregate HMIS reporting via district** ← likely the majority
2. No local DHIS2 access, reports on paper to an intermediary
3. Reports directly but **late**
4. Reports but **incomplete** (subset of indicators)
5. Reports **inconsistently** (some months)

6. Genuinely absent from the national feed

Why this changes things materially: if most of the 175 do reach DHIS2 as monthly aggregates, then (a) the bulletin's coverage is far better than I assumed and the sampling-bias warning in 2.4.1 becomes conditional rather than certain, and (b) the argument against B shifts, B fails not on *absence* but on *granularity and freshness*, since monthly aggregates cannot drive real-time facility status regardless of how many facilities send them.

1.4 Patterns I would hunt

- **Re-keying points.** Every place a human retypes data between systems. Latency source, defect source, and automation candidate simultaneously. Count and time them.
- **Shadow spreadsheets.** Every emailed Excel file is a system failing someone. The brief already names one (immunisation).
- **The reconciliation tell.** No answer to "what happens when sources disagree" means nobody is checking, that is the mess.
- **Decision latency vs. data latency.** *Decisive for B.* If districts decide monthly or quarterly, 3-week-old data is not the binding constraint.
- **The denominator problem.** Do current catchment populations exist? Every *rate* depends on them, and a wrong denominator is invisible.
- **Accountability mapping.** Data quality improves only where someone is measured on it.
- **Definition drift.** Do "ANC visit" and "complication" mean the same thing in HealthTrack, OpenMRS and DHIS2? If not, the bulletin aggregates incomparable things.
- **Who loses from automation.** The analyst's 40 hours are someone's job content. Champion or blocker? This determines whether A's political theory holds at all.

1.5 What I would observe directly, not take on report

1. **Sit through a bulletin build with a stopwatch**, screen-recorded with permission. The critical measurement is the **mechanical : judgement ratio**, it sets the automation ceiling.
2. **Open the actual DHIS2 export.** Completeness per district, missing org units, duplicate submissions, how late the late ones are, and **when in the cycle the data actually lands**.
3. **Attempt to reproduce last quarter's published figures from source.** If I cannot, the correct conclusion is "*this is not independently reproducible from the materials supplied*", not "nobody can." The analyst may hold undocumented exclusions and corrections, and finding those is the point.
4. **Watch month-end at a paper-only facility**, the register, the transcription, the person, the light. This is also how 1.3 gets answered.
5. **Watch a district officer make a real decision.** Not describe one.
6. **Measure the infrastructure claim.** "4–6 hrs/day power, spotty 3G" is inherited from the brief. Verify where it matters.

1.6 What would change my mind

OBSERVATION	FALSIFIES	RESPONSE
Districts decide **weekly** and cannot today	The case against B	Re-scope toward B for facilities with adequate granularity, honestly named as partial
The 40 hrs is mostly **judgement**	The ROI case for A	Automate assembly only; reset the number publicly
The bulletin is **not read** by any decision-maker	A's value entirely	Stop. Find what *is* read
Co-infected patients harmed **today at measurable volume**	The deferral of C	Escalate as clinical safety, resource properly
Hosting/InfoSec approval takes **> 3 weeks**	The whole sprint shape	Re-plan around an offline deliverable; escalate on Day 3

Section 2, Problem Selection

2.1 A, B and C are solutions, not problems

The brief states three *things to build*. Mapping each back to the need it serves surfaces siblings the brief omits, and one need it misses entirely. The statements below are my hypotheses phrased in stakeholder language, not interview evidence: they are what I would try to falsify in Week 1, not things anyone has said.

Root outcome: MoH leadership acts on health-system data it can defend, early enough to change the next planning cycle.

NEED	WHOSE	THE BRIEF'S ANSWER	SIBLINGS WORTH PRICING
O1 "I spend most of a working month on a document that is stale before it is read, and I cannot defend its numbers."	MoH analyst	**A**, automate the bulletin end to end	Shorten the cadence, quarterly → monthly; retire the PDF for a live page
O2 "By the time I see a stockout it is a crisis, never a warning."	District Health Officer	**B**, real-time facility dashboard	SMS exception reporting from facilities; supply-chain-only feed (eLMIS), no clinical
O3 "Co-infected patients are counted twice and treated once."	TB + HIV programme managers	**C**, unified patient view across CommCare	**Weekly reconciliation report, no merging**; shared patient identifier (policy, not software)
O4 "I cannot defend the numbers I am shown, so I do not act on them."	MoH Director	**none, not in the brief**	Every figure traceable to its DHIS2 value + snapshot; completeness published beside every indicator; a past quarter reproduced from source as a trust proof

Two things the A/B/C framing hides. O4 may be the Director's actual complaint, and none of A, B or C answers it, because all three answer throughput. Whatever ships should therefore carry traceability and published completeness as requirements rather than features. Separately, the weekly reconciliation report is a cheaper answer to O3 than C: most of the value, none of the merge risk, and the brief does not consider it. Both are hypotheses to test in Week 1, not findings.

2.2 The choice: Problem A, plus one bounded handover act

> **Commitment (conditional on the Week 1 gate in 2.5a):** automate the Quarterly Health Bulletin, with figure-level lineage built in from the start, and one named Digital Health Officer independently producing a bulletin from a replayed quarter before I leave.

What I am giving up, stated first. The highest-value outcome for this Ministry is probably operational, closer to O2 than O1. A does not deliver that. I am choosing the reachable wedge over the valuable-but-unreachable one, and the honest framing to the Director is "*this is the thing I can finish and prove in six weeks; here is what it does not do*." Selling A as though it serves the operational need would be the failure mode.

Why A. It is the only option that can finish and be proven inside six weeks, operating on data that already arrives in a shape that already exists. It is the only one with a baseline that plausibly already exists, subject to **A3**, which must be measured rather than assumed. And it is the trust wedge: a defensible artifact delivered to the analyst and the Director buys the standing to attempt B, and in a first-country, template-setting engagement, proof is the currency.

Why not B, why not C. B is out of reach *in this sprint*, not impossible: it needs facility-level freshness, and the reporting substrate is monthly aggregates on a 2–3 week lag. That is a granularity and cadence gap rather than a coverage gap (1.3), and closing it is a separate project. C matches records across CommCare silos with no shared patient identifier, so matching is probabilistic in both directions. A wrong match puts one patient's HIV status into another's treatment decision, which matters concretely: rifampicin, first-line for TB, collapses the plasma levels of several antiretrovirals, so co-infection status drives the regimen. A missed match loses the co-infection entirely. A unified *view* does not itself prescribe treatment, so the risk turns on whether the match is advisory or authoritative, and settling that needs a named clinical reviewer and a Ministry sign-off route that do not exist here and cannot be created in six weeks. The weekly reconciliation report is the version of C I would propose: it flags a likely co-infection for a person to check and merges nothing, so the system makes no clinical claim.

What A does not give B, since the sequencing pitch is where this gets oversold. A builds an aggregate, period-grain mart on bulletin cadence; B needs facility-status freshness, stock and supply events, intra-period updates, alert state and freshness monitoring. What genuinely transfers is the **org-unit dimension and the indicator dictionary**: real, reusable, and far less than a foundation. Two further claims I will not make: that anything here is greenfield-free, since Superset, dbt and Airflow appearing in Sand's job postings evidences the component types, not a working Rwanda-configured instance (gate **G4**); and that the bulletin is necessarily templatable, since a government quarterly is typically narrative + tables + interpretation rather than a dashboard export, and **A5** resolves whether I am automating the Ministry's artifact or quietly redefining it.

The internal disagreement I would surface, not hide. The Solutions Manager pre-identified the use cases before discovery, and his hypothesis points at B. Discovery points at A. I would put the sequencing in writing to him and the Director in Week 1, including the honest version of what A does and does not give B. An FDE who lets a pre-sold narrative survive contradicting evidence has chosen internal comfort over the client.

2.3 The measurable outcome

Measured from **data availability**, not from quarter close. DHIS2 runs 2–3 weeks behind and nothing in A makes inputs arrive faster; generating a document faster does not make its data appear, and publishing fast on incomplete data would *reduce* trust, which is the opposite of the goal.

MEASURE	BASELINE	TARGET
Analyst time per publication cycle	**To be measured in Week 1** (see below)	≥ 90% reduction
Data-availability cutoff → published	~1–2 weeks after data lands	**≤ 2 working days**
Published figures resolving to a DHIS2 value + snapshot	0%	100%
Facilities with reporting status shown	not shown	100%, by the six states in 1.3
Ministry can produce a bulletin unaided (replay test)	no	yes

On the baseline number. The brief says "*40 hours/month*" for a *quarterly* bulletin. Those are inconsistent by roughly 3×: it could mean 40 hrs every month on reporting generally, 40 hrs in the publication month, or ~120 hrs per quarterly cycle. **I commit to a 90% reduction, not to a number, until the stopwatch resolves it in Week 1.** Committing "40 hours → 30 minutes" before measuring is how a credibility problem gets manufactured in Week 2.

On lineage. Traceability is to *the DHIS2 data value, org unit, period, version and extraction snapshot*, not to the underlying facility register or patient record. For paper-sourced aggregates, record-level provenance does not exist to trace to.

One adoption measure, because hours saved is production efficiency, not health value: is the bulletin reviewed before a named decision meeting, and does any exception in it generate an assigned action? If A meets every engineering metric and this stays "no," A has not worked.

2.4 Explicitly out of scope

- **HealthTrack EMR integration** (45 hospitals, buggy, local servers). Its own project.
- **Problem C in any form**, including the reconciliation report: proposed, not built.
- **Real-time anything.** That is B.
- **Correcting source data.** We surface completeness; we do not fix DHIS2's contents.
- **Patient-level data.** Aggregate only.
- **Redefining any indicator.** A Ministry decision, and the classic scope-creep vector on reporting projects.
- **New hardware, new mobile apps, new user accounts** outside the existing bulletin workflow.

Two things that look out of scope and are not, because unresolved they consume the sprint. **Hosting, service accounts and InfoSec approval:** where the pipeline runs for MoH Rwanda, who holds credentials after Week 6, and whether national health data may sit in a Sand-hosted environment. Day 1 question, Day 3 escalation. **Small-cell suppression:** "aggregate-only, therefore no PHI" is too categorical, since rare events at facility level combined with geography can re-identify. A suppression rule is required before publication, and access control, audit logging and retention remain in scope regardless.

2.4.1 The 175 paper facilities

Out of scope for six weeks, and the largest thing this bulletin will not see. How many of the 175 already reach DHIS2 as monthly aggregates via district clerks is unestablished (1.3), so the size of the gap is unknown, and whatever it turns out to be, the bulletin should print it: paper-only facilities are plausibly smaller and more rural, which is where maternal and neonatal mortality is likely highest, so a "top 10 facilities by volume" computed over reporting facilities only would systematically describe the better-resourced end of the system. *I am asserting that, not evidencing it; it is checkable against the provided facility data and should be checked.* Closing the gap is a measurement programme rather than a pipeline, and the experiment that decides it is cheap: ~50 real register photographs, extraction run, field-by-field accuracy against manual transcription. Costed in the repository, not here.

2.5 Assumptions and Week 2 validation

#	ASSUMPTION	VALIDATION	IF FALSE
A3	The reporting labour is mostly *mechanical*	**Stopwatch**, Week 1	Automate assembly only; reset the target publicly
A4	DHIS2 API access granted to a service account	Ask Day 1, escalate Day 3	CSV export drop, same mart, same output
A5	The bulletin is reproducible as a templated artifact	Inspect last 2 editions, Day 2	Generate tables/charts only; narrative stays manual and is named as such
A6	≥2 machine-readable historical editions exist	Request Week 1	Ship without trends in v1
A7	A **named** DHO with cleared hours exists	**Written ask, Week 1 Day 2**	Downgrade honestly; stop calling it capacity transfer
A8	The bulletin is read by someone who decides	Ask Director + 2 district officers what they did with the last one	Stop and re-scope
A9	Indicator definitions are consistent across sources	Sample-compare across 3 facilities	Scope to consistent indicators, flag the rest
A10	Hosting and InfoSec approval achievable in ≤ 3 weeks	Meet MoH ICT Day 4	Re-plan around an artifact that runs on Ministry-owned infrastructure
A11	A quarter close or a replayable prior quarter falls inside the sprint	Check the reporting calendar Day 1	Use the replay protocol in 2.6

2.5a The Week 1 gate

Committing at the end of Week 1 while planning to resolve existential assumptions in Week 2 is incoherent: **A8** alone can invalidate the whole commitment. The commitment in 2.2 is therefore **conditional on five gates**, assessed at end of Week 1.

GATE	PASSES IF
G1, Value	At least one named decision-maker used the last bulletin for something identifiable
G2, Mechanisability	≥ 60% of measured cycle time is mechanical assembly
G3, Inputs	DHIS2 access (API or scheduled export) confirmed, with a stated availability cutoff
G4, Runtime	A viable place to run and hand over the pipeline is identified and approvable

GATE	PASSES IF
G5, Operator	A named person is assigned to receive it

G1 or G3 fails → A is the wrong commitment, and I say so in Week 1 rather than Week 5. **G2 fails → the outcome is re-stated smaller** before anything is promised. **G4 or G5 fails → build proceeds, handover claim is dropped**, honestly and in writing.

2.6 Success metric

Primary: percentage reduction in analyst time per cycle, target ≥ 90%, against a Week-1 measured baseline. **Secondary:** data-availability → publication, ≤ 2 working days; figures resolving to a DHIS2 value + snapshot, 100%. **The one that matters:** *can the Ministry produce it without me?*

Because a Q[N+1] edition may not fall inside a six-week window (**A11**), that is tested by **controlled replay** in Week 6 rather than left to luck. The named DHO, unaided, triggers the pipeline for a prior quarter, diagnoses a **seeded** failure (a deliberately broken credential or a malformed export), reviews the completeness exceptions, generates the bulletin, routes it for the normal approval step, and publishes it. The seeded failure and the approval step are the two that usually get skipped, and they are the two that actually predict survival.

The Week 3 demo is not a dashboard tour. It is: "*here is last quarter's published bulletin, here is the pipeline re-deriving it from source, here are the figures that match, and here is the one that doesn't, and why.*" A reconciliation that surfaces a genuine discrepancy is a far stronger trust event than one that finds none.

Ownership after exit needs naming, not just a person: who owns pipeline operations, indicator definitions, source corrections, late submissions, bulletin approval, and infrastructure incidents. Without that, every anomaly routes back to Sand, or the bulletin stalls between generation and approval.

2.7 Fallback plan

Each assumption in 2.5 has a stated failure branch, and none ends the engagement. A blocked DHIS2 API (**A4**) becomes a scheduled CSV export into the same mart, costing ingest automation and nothing else. A bulletin that is not templatable (**A5**) means automating tables and charts while the narrative stays manual and labelled. Contested definitions (**A9**) ship the uncontested sections and annex the rest against a named Ministry decision; sparse history (**A6**) ships the current quarter with a completeness report. No named DHO (**A7**) ships the artifact correctly labelled with the capacity-transfer claim dropped; blocked hosting (**A10**) delivers mart, queries and runbook on Ministry infrastructure with no Sand dependency; labour that proves mostly judgement (**A3**) resets the target publicly in Week 2, a smaller true claim instead of a larger false one.

The floor, if everything slips: a conformed mart, a documented query set and a runbook. Even at worst that removes the re-keying step, is handoverable, and starts the next engagement from a warehouse rather than zero. The branch-by-branch table is in the repository.

Deliverable 2, Solution Design and Implementation Notes

Written counterpart to `../artifacts/06-bulletin-architecture-data-flow.html` (mart ERD as inline SVG). This document carries the required prose (products, what is custom, build vs. buy) and stands alone when printed. Section 2 (prototype code) is done: see `README.md`. Covers Section 1 in full, Section 3 where the README does not already.

Section 1: Solution Design and Architecture

1.1 Architecture and data flow

Four layers, plus checks, plus render. Python stops at Parquet (ADR 0010); everything past that is Astro.

Sources. The assignment's five CSVs (`facilities`, `clinical_neonatal`, `governance`, `healthcare_workers`, `operations`), plus `dhis2_sample.csv`, a DHIS2-shaped source, hand-built to prove the crosswalk pattern, not a live API pull (see 3.1).

Bronze (`dataflow/bronze.py`). Loads all six sources verbatim, `dtype=str`, stamped with `_source_system`, `_source_file`, `_source_row`, `_batch`, `_row_hash`, `_ingested_at`. Where facility and period arrive twice (2024-01 and 2024-03, 234 rows), both loads are kept: silently colliding and picking a winner is the defect this stamps against.

Crosswalk and identity (`mart/crosswalk.csv`, `mart/org_unit_map.csv`). Two declared CSV tables, not code. The crosswalk maps 71 source fields (66 assignment, 5 DHIS2) to a canonical element, role (`identity`, `dimension`, `observation`, `unmapped`), and note; `org_unit_map` declares which facility identifiers across source systems mean the same facility. Both read as data at runtime, not hardcoded (see 1.4).

Silver (`dataflow/silver.py`). Melts every wide source into long form, unions it with the DHIS2 source, inner-joins to the crosswalk (dropping every `unmapped` field, see 3.1 on GPS), resolves identity through `org_unit_map` (raising on an unresolved key, not inventing a facility), and emits one canonical `observation` at DHIS2 grain: `(org_unit, period, data_element, batch)`.

Gold (`dataflow/gold.py`). The marts a bulletin actually reads: `observations_resolved` (batch collision resolved, arbitrarily, see 3.1), `facility_quarter / district_quarter`, `nmr_facility_quarter / nmr_district_quarter` (neonatal mortality rate per 1,000 live births, WHO GHO definition), `completeness_summary`, `facility_capability / capability_summary`, plus two declared-not-derived tables read verbatim: `known_contradictions.csv` (8 findings, each a contradiction between two populated fields, not missing data; see 3.2 for an example) and `cause_capability_links.csv` (4 cause-of-death to capability pairings).

Checks (`dataflow/checks.py`). Two statistical tests, seeded (`PERMUTATION_SEED = 20260810`, 200 trials), that annotate a claim rather than gate a row: `temporal_signal_check` (momentum vs. a shuffled baseline) and `stratification_check` (pooled correlation split by facility tier). Both always emit a caveat; ADR 0012 made a failed check annotate rather than hide the section it qualifies.

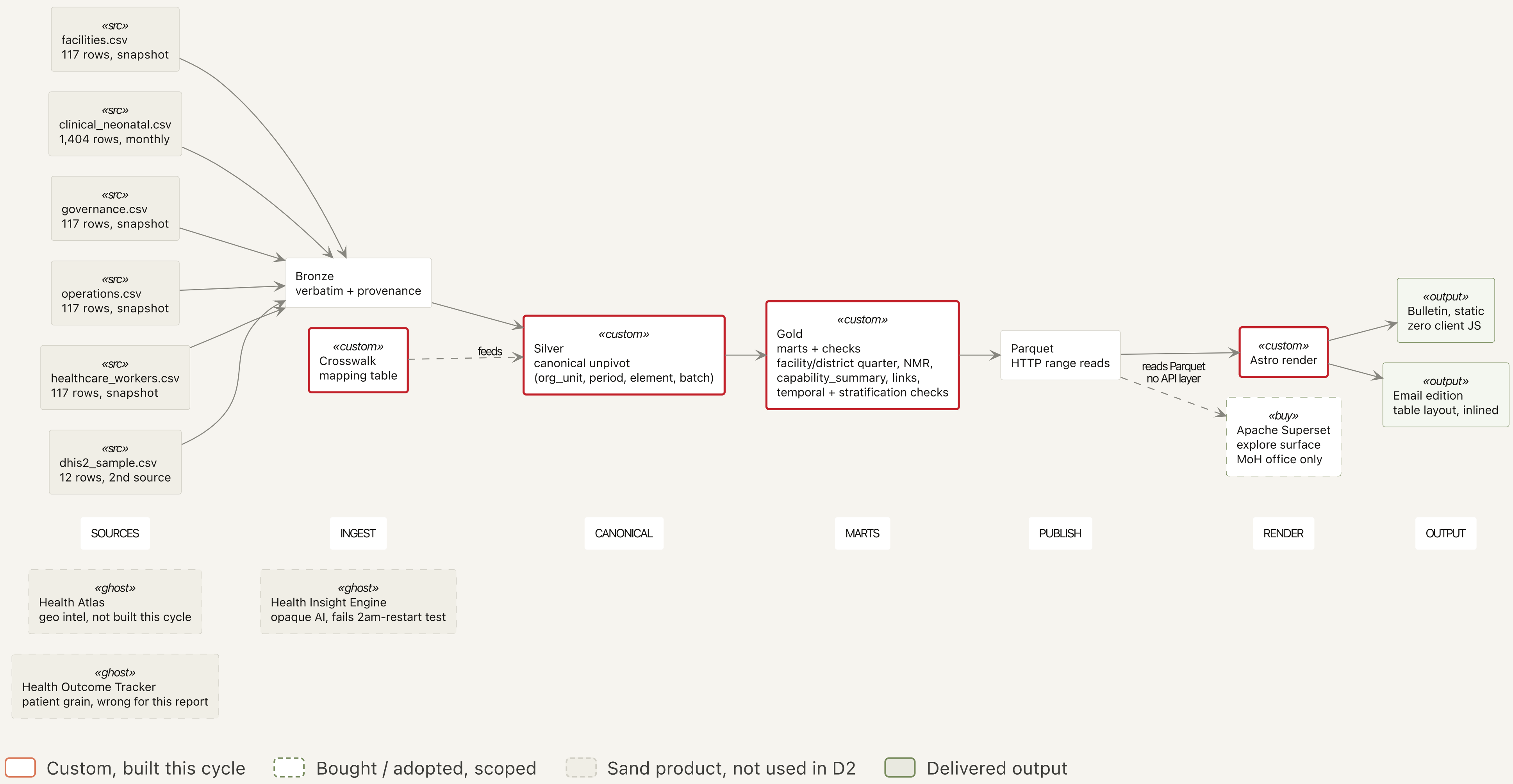
Publish, then render. All of the above lands in `mart.duckdb` and individual `.parquet` files. Astro reads Parquet at build time and renders three surfaces from one component set: the static Bulletin (zero JavaScript, one file per quarter, works forwarded and offline), the Email edition (a separate, smaller document; see 1.3), and an Explore surface (Mosaic over DuckDB-WASM), scoped but not built (see 3.2). Apache Superset also reads the same Gold-layer Parquet, server-rendered, scoped to the connected central MoH office only (1.2, 1.3).

The rendered version of this diagram, plus the mart ERD — deliberately not a star or snowflake schema — are reproduced from `artifacts/06-bulletin-architecture-data-flow.html` in the PDF submission; open that file directly for the interactive version, including the product-decision cards (1.2 below covers the same ground in prose).

1.2 Which Sand products, and why

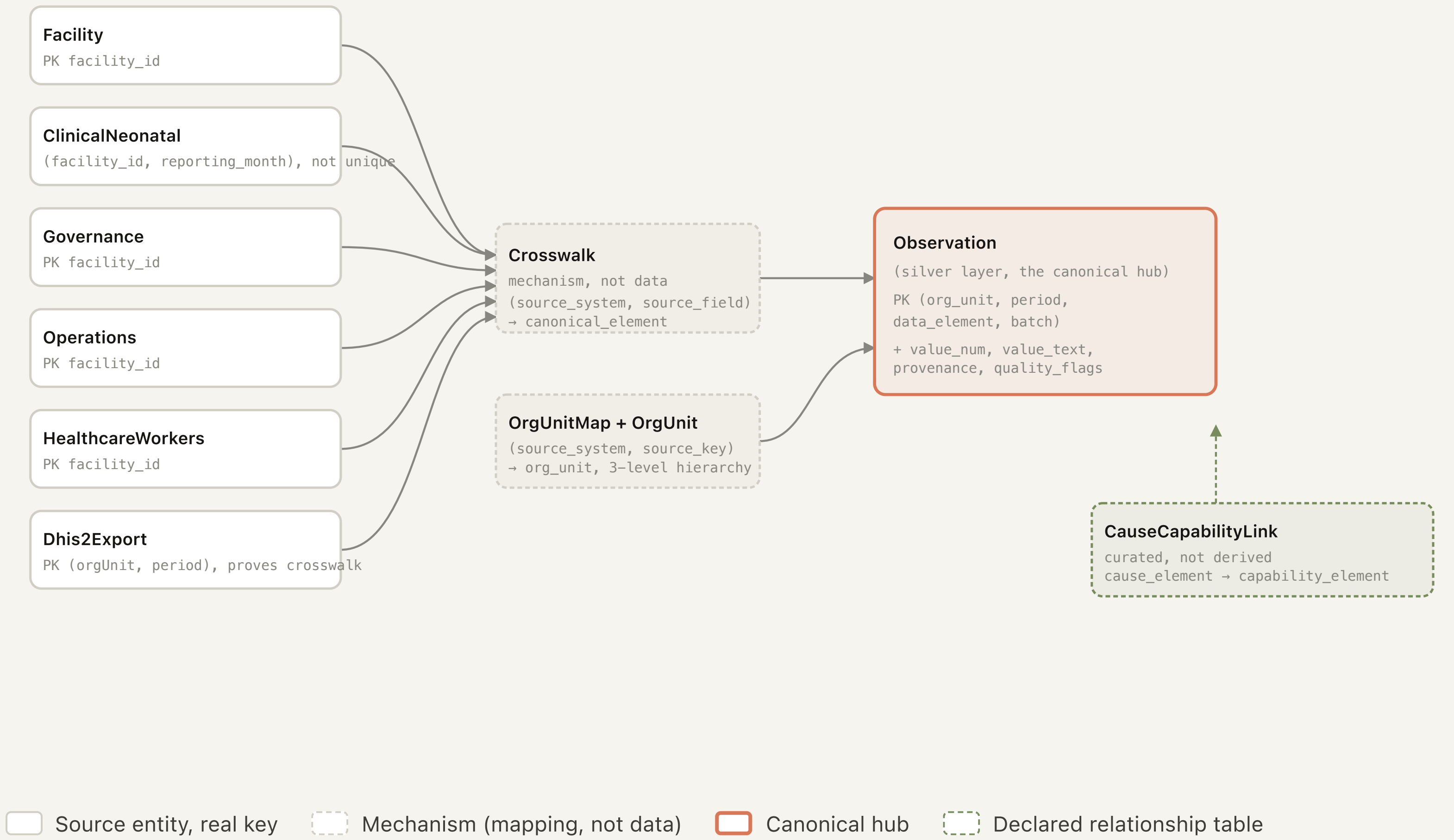
Firm honesty constraint, applied throughout. Sand's stack components (Superset, DHIS2 integration, dbt/Airflow) appear in Sand's own FDE job postings. `research/sand-product-research.md` grades that evidence *strong*, not *confirmed*: good evidence the component types exist and are staffed for, not evidence a working, Rwanda-configured instance exists today. D1 made exactly this error once (job-posting copy read as existence proof), caught in review and downgraded to a Week 1 question (gate G4). Below, "in Sand's stack" and "confirmed live for this engagement" stay two separate claims; the second is never asserted without a source that supports it.

PRODUCT	USE IN THIS DESIGN	WHY	EVIDENCE GRADE
Analytics Template Toolkit (Apache Superset)	Yes, scoped to the connected central MoH office audience only	Superset is server-rendered, needing a live connection: fails in low-power districts (4-6 hrs/day power, spotty 3G/4G) but works in the central office. The Gold-layer marts are already Superset-ready: documented Parquet/DuckDB tables with a field dictionary in `crosswalk.csv`. Two audiences, two tools, one data model (ADR 0008).	Moderate: named directly in a Sand FDE job posting (Nigeria), corroborated by a second employee's LinkedIn profile (URL not captured). A live, Rwanda-configured instance is unconfirmed; per D1's gate G4, a Week 1 question, not an assumption here.
HealthOS Data Models	No	This binds directly to DHIS2's own public data model (`org_unit`, `period`, `data_element`, ADR 0009): the naming conflict resolved (Bluelake's UI vs. the CSVs' fields vs. DHIS2's UIDs vs. the brief's prose) is specific to this engagement's sources. A generic product cannot pre-resolve a conflict defined by which systems a Ministry runs and what its analysts call things.	Weakest of the five. The research doc's entire "likely architecture" section for this product is marked `[INFERENCE]`, drawn from generic DHIS2/openHIE/medallion patterns, not from anything Sand-specific.
Health Outcome Tracker	No	The bulletin's facility to district to national indicator roll-up is built directly in the Gold layer (`district_quarter`, `nmr_district_quarter`). No working, Rwanda-configured instance is established to plug a new bulletin into. If one exists, in a real engagement the roll-up logic would migrate to configuration on that product rather than stay bespoke Hamilton nodes (see 1.3, cost to reverse).	Moderate: the pillar name and scope trace to an AWS Marketplace listing (the research's strongest technical source); architecture inferred from one employee's LinkedIn profile; quantitative outcome claims (ANC4 increase, etc.) are vendor marketing, not audited.



Synthetic test data throughout (assignment CSVs plus a hand-built DHIS2 sample) — not real Rwanda health records.

Not a normalized relational schema. There is no web of pairwise foreign-key joins here. The real shape is six heterogeneous source tables, each with its own key and grain, unpivoted through one crosswalk mapping into a single canonical long-format fact table. Drawing this as a conventional star or snowflake ERD would misrepresent the architecture, so it is drawn as what it actually is: sources feeding a mapping mechanism feeding one hub.



Synthetic test data throughout (assignment CSVs plus a hand-built DHIS2 sample) — not real Rwanda health records.

PRODUCT	USE IN THIS DESIGN	WHY	EVIDENCE GRADE
Health Insight Engine	No	Documented shape: AI-driven anomaly detection and alerting. The closest analog, Rwanda's National Health Intelligence Center, is live (2026 outbreak surveillance), but the brief's four metrics (top facilities, maternal indicators, performance scores, trend) are descriptive analytics, not alerting: no fit even where a confirmed instance exists. The two checks built instead are simple, seeded, auditable, the opposite of an opaque model: a DHO needs a traceable method, not a score to trust unread.	Best-evidenced architecture of the five (Rwanda MoH published its own six-layer HIC diagram), but the AI/alerting mechanics themselves remain unverified inference even against that source.
Health Atlas	No	GPS coordinates are fabricated (uniform random inside Rwanda's bounding box, every province spans the full extent), structurally excluded at the crosswalk's `unmapped` role, not just hidden downstream: no facility-point layer to hand to a mapping product. If real coordinates arrive later (Deliverable 3 proposes HDX's Rwanda Healthsites layer, 1,345 facilities), this reopens, including whether Health Atlas is the right home.	Weakest single sourced fact of the five: the 3D-mapping capability traces to one LinkedIn post by a named engineer describing Rwanda work; ingestion and use details are inference from cross-vertical Sand language.

Net: one product used (Superset), four not, each for a stated evidence-or-fit reason. None of the four "not used" calls says the underlying capability is fictional, only that this design does not get to assume it is available.

1.3 Build vs. buy, per component

COMPONENT	DECISION	WHY	COST TO REVERSE
Ingestion (DHIS2 and legacy CSV sources)	Build: `crosswalk.csv` + Hamilton loaders in `bronze.py`	No evidence a Rwanda-configured DHIS2 connector exists in Sand's stack (D1 gate G4); a reviewable CSV fits the "restart alone, 2am, no internet" test better than a connector's internal mapping (ADR 0006, 0008).	Low to moderate: `crosswalk.csv` grows by adding rows; a real connector would replace `bronze_dhis2` only, since silver and gold already read a crosswalk-mapped table.
Transformation and mart (bronze to silver to gold)	Build: Apache Hamilton + DuckDB + Parquet	Passes the dependency test the stack is chosen against. DAG derives from function parameter names, so lineage comes from code. Server-side today; a Pyodide/browser path is asserted from `tryhamilton.dev`, unverified here (ADR 0008).	Moderate: Hamilton nodes are backend-agnostic pandas/pyarrow functions; swapping DuckDB for a warehouse re-points `run.py`, not the DAG.
Orchestration and scheduling	Neither built nor bought yet: `run.py` is invoked by hand today	A quarterly batch completing in seconds doesn't clear the bar for a durable-execution runtime; DBOS was rejected on the same test (ADR 0008). Nothing runs on schedule, no failure alerts (Deliverable 3, item 4).	Low: Hamilton exposes the DAG as a callable graph; wrapping it in cron or the Ministry's own scheduler is additive.

COMPONENT	DECISION	WHY	COST TO REVERSE
Chart rendering	Buy: Flint (`microsoft/flint-chart`, npm), over hand-rolled Vega-Lite or D3	One spec compiles to a static render (email, build-time SVG) and an interactive backend (unbuilt explore surface). Constrained grammar and MCP server make agent-authored charts validatable; raw Vega-Lite or D3 would not be (ADR 0008). Deferred in ADR 0010 over npm-only packaging while Python-rendered, re-adopted in ADR 0012 once Astro/TypeScript.	Moderate: a real Vega-Lite compiler bug was worked around (custom sort dropped under multi-layer auto-labels), unfiled upstream. Two details (a hatch fill, a benchmark rule line) use a hand-edit escape hatch since Flint's semantic layer doesn't expose them.
Document rendering (bulletin and email)	Build: Astro, zero JavaScript by default, over a Superset export or the earlier hand-rolled Python renderer (`render.py`, deleted)	Email clients strip script, so a Superset export or SPA cannot produce the email surface at all. The earlier hand-rolled Python renderer duplicated logic against the interactive surface, capping UX (ADR 0010).	Low to moderate: Astro components are the one thing every surface shares; replacing the renderer rebuilds all three, but the Parquet contract doesn't change.
Explore and dashboard surface, connected office	Buy: Apache Superset	Server-rendered dashboards over a live connection suit reliable power and connectivity; not attempted for the offline majority, where it would fail.	Low if a live instance exists in Sand's stack (unconfirmed, see 1.2); moderate to high if stood up from scratch: hosting, credentials, InfoSec approval all open (D1 2.2).
Distribution (email)	Build: a deliberately separate, smaller HTML document, not the bulletin inlined	Measured, not assumed: inlining the full bulletin hit 99.0 KB against Gmail's ~102 KB clip threshold, with trust-critical disclosure sections (withheld panels, known defects, lineage) at the bottom, first clipped. Inline SVG is also unsupported across major clients (ADR 0011).	Low: already a distinct document; reversing to inline-everything is a template change that reopens the clipping problem ADR 0011 measured.
Hosting	Deferred: no hosting decision made, none needed for a local prototype	D1 originally assumed Sand-hosting, flipped to explicit in-scope on review (2.2). Where this runs, who owns credentials after week 6, and whether health data may sit in a Sand-hosted environment are open, not settled by a laptop prototype.	Unknown, deliberately unestimated: deciding without InfoSec or Ministry input is the same Rwanda-specific-instance assumption D1 already caught once.
Statistical checks (temporal signal, stratification survival)	Build: permutation test and stratified correlation, in-repo, seeded	Tests this bulletin's claims against this dataset's structure (117 facilities, 4 tiers, quarterly). None of the five Sand products does statistics this in-context; Health Insight Engine's alerting is closest, unverified, and different (anomaly detection, not per-claim validity).	Low mechanically: ~150 lines of pandas/numpy, seeded, no dependency. The real cost is institutional: a Ministry reading a caveat as correct, not a bug.

COMPONENT	DECISION	WHY	COST TO REVERSE
Facility identity resolution	Build: declared `org_unit_map.csv`, not auto-matching on a shared key string	Two systems using `NYA001` for the same facility is evidence of a shared code space, not proof of identity. Auto-matching on string equality is how an identity graph silently fuses two real facilities.	Low: additive rows; the real cost is process, since someone must make and record each match.

1.4 What is custom, and why

Three things are genuinely custom (written for this engagement, not configuration on a bought tool), each for a specific reason, not by default.

The crosswalk and indicator dictionary (`mart/crosswalk.csv`). This resolves a real naming conflict: Bluelake's UI says "Health Post" in chart titles and "Facility" in its filter bar; the sample CSVs say `facility_id`; DHIS2 says `orgUnit`; the brief says "facility" and "site" interchangeably (ADR 0009). That conflict is specific to which systems this Ministry runs and what its analysts call things. A bought mapping tool resolves a generic version; it cannot resolve this one without the same 71-row harvest already done, because the harvest is the work.

Per-figure lineage. Carried as columns through the pipeline (`rules_applied`, `quality_flags`, `provisional`, `source_system`, `source_row`, `row_hash`, `ingested_at`), not a bolted-on audit log. `observations_resolved` publishes to Parquet because it is the only table carrying `rules_applied`, so the bulletin's lineage section reads the actual rule name, not a restatement. A generic BI tool's lineage feature describes a generic ETL's; it cannot name this pipeline's own arbitrary tie-break (`DEFAULT-BATCH-01`, see 3.1) without this pipeline's code producing that fact first.

The two statistical checks. As in 1.3: they answer whether a specific correlation survives being split by a specific stratification variable, narrower than the question a bought anomaly-detection layer is built to ask ("is this point unusual").

One non-custom call on the opposite principle: `known_contradictions.csv` and `cause_capability_links.csv` stay declared CSV data, not code. Pairing a cause of death with the capability that treats it, or flagging a kangaroo-care contradiction, is a clinical judgement call, not a statistical inference; encoding it in Python would make it harder to review, the same failure mode the crosswalk avoids.

Section 3: Implementation Notes

3.1 Shortcuts taken

`README.md` already lists four (lineage as file anchors rather than addressable URLs, the `DEFAULT-BATCH-01` arbitrary tie-break, gold regenerating fully on every run, and the stratification check testing tier only). This adds the ones that document does not cover.

- **Second source is synthetic, not a live DHIS2 pull** (see 1.1). `bronze_dhis2` loads a hand-built, 12-row, 3-facility CSV shaped like a DHIS2 export, to prove the crosswalk's claim that a new source onboards via mapping rows, not pipeline code; not evidence of a real API pull.
- **No scheduler.** The full build (`uv run python run.py && bun run publish`) runs by hand. Nothing runs on a quarterly cadence and no one is told when a run fails. Hamilton already exposes the DAG as a callable, so this is a wrapper away, not built yet (Deliverable 3, item 4).

- **No auth.** No server process, nothing to authenticate against: no API key, no cloud account, no login. A consequence of the offline-first design, not deferred; the day this runs against a live source or a hosted surface, auth becomes a real open question untouched here.
- **GPS excluded because it is fabricated, not out of scope to try** (see 1.2). Coordinates are structurally dropped at the crosswalk's `unmapped` role before silver, not just hidden in the render. The bulletin maps at district grain, the coarsest the real data supports.
- **Single-machine DuckDB.** `mart.duckdb` is a local file, no server, no connection string. Correct for a quarterly batch running in seconds; ADR 0008 names its own reversal condition (growing long-running or multi-stage enough to genuinely fail partway), which has not happened.

3.2 What I would show in Week 3, what works, and what is genuinely broken

D1 2.6 specifies the Week 3 demo as a reconciliation, not a dashboard tour: last quarter's published bulletin, the pipeline re-deriving it from source, the figures that match, and the one that does not, with why. This build cannot do that literally: the data is synthetic and no prior bulletin exists to reconcile against, a gap between plan and what is buildable here, named rather than glossed over.

What works today. All four 2024 quarterly editions build and pass the five checks (`check-shipped` , `check-tokens` , `check-style` , `check-email` , `check-agreement`), plus the publish-time guard that refuses to write a file whose name and contents disagree on the quarter (shipped once with a mismatched filename, hence the guard). Of the four required bulletin metrics, two are answered directly and two are substituted with the substitution declared: top-10 facilities by volume, and a genuine four-quarter trend rather than a two-point delta, are direct. Maternal health indicators are not: the provided CSVs carry no ANC column and no complication column, so what ships is neonatal cause of death and the stillbirth ratio, which are perinatal rather than maternal. Facility performance ships completeness but not timeliness, because no source file carries a submission timestamp, so no lag is derivable. Both gaps are in the extract, not the design, and both close with three more crosswalk rows against standard DHIS2 elements in Week 2. The full accounting is in `README.md` . The checks disclose rather than hide: a correlation that fails stratification, or a trend with no measured temporal signal, ships with a caveat sentence instead of an empty "withheld" section (ADR 0012). Eight known contradictions surface by name, not corrected silently (`staff_last_training` = "Never" at facilities that also report trained nurses, a referral imbalance, oxygen plants without backup power, among others).

What I would demo instead: the pipeline re-deriving all four quarters from the same six source files live, then walking one `known_contradictions` row end to end, from the raw CSV rows through the crosswalk to the exact sentence in the bulletin, so the audience sees a real discrepancy traced through every layer, not asserted. Same trust event D1 2.6 is after (a genuine mismatch, explained, not a clean run with nothing to show), built from what this pipeline actually has.

What is genuinely broken, beyond the reconciliation gap above.

- The batch conflict has no analyst-actionable resolution: disclosed (every affected figure marked provisional) but not fixable in-tool; the triage queue is specified in `openspec/changes/data-validation` , not built.
- Ingestion assumes one file shape per source, learned by hand. Real DHIS2 exports, HealthTrack dumps, OpenMRS extracts, and paper forms will not hold still, and the mapping table is a CSV a human edits, not a confirmed pipeline.
- The explore surface (architecture diagram, ADR 0012's honest-limits note) does not exist; chart-runtime is open for whatever gets built there.

- `pipeline/README.md` is empty, the file someone would open first.
- The stratification check tests facility tier only, not a declared variable per association, though its design implies it should.

3.3 What I learned from building this

The clearest finding is a confound, already shipped in the bulletin, not found after the fact for this document. `checks.stratification_check` computes, and `mart/stratification_check.parquet` records: pooled correlation between `governance_staff_trained_rate` and neonatal mortality across all 117 facilities is **r = -0.844**. Split by facility tier, it flips sign and collapses toward zero in every tier: **District +0.111, Health Center +0.042, Provincial +0.144**. The pipeline's own caveat states it directly: "the association describes tier, not the covariate." A second covariate, `capability_cpap_machines`, shows the same shape pooled (**r = -0.811**), failing harder within tier (only Provincial has enough variance to test, **-0.269**). A bulletin reporting either pooled figure alone would tell a Ministry that training or an equipment purchase drives survival, when what actually drives both the correlation and the outcome is which tier of facility a patient reaches: District and Health Center facilities have worse neonatal outcomes and less of nearly everything, independent of any specific covariate.

Section 6 of the published bulletin also reports a composite equipment index, built in the web layer, not `checks.py`: pooled **r = -0.867** across all 117 facilities (stated in prose as -0.87), collapsing within tier to **District -0.071, Health Center +0.036, Provincial -0.288**. I verified this independently: a z-scored sum of nine equipment fields against average facility-level mortality gives pooled **r = -0.868** (n = 117), within tier District **-0.06**, Health Center **+0.04**, Provincial **-0.26** (n = 45, 45, 25; Referral excluded), deferring to the bulletin's published figures rather than competing with them.

The two constructions agree in sign and land within a hundredth pooled, so the pooled effect is robust to which fields get chosen. The within-tier numbers are not: they move by two to four hundredths between constructions at n=45 (District, Health Center) or n=25 (Provincial) per tier, small enough that composing the index differently (which fields, how weighted) can shift it. That instability is the finding: once the tier effect is removed, what remains is too small and too dependent on construction choices to support a claim either way. A bulletin publishing a within-tier number as stable and composition-independent would be making the same overclaim the pooled figure makes, one level down.

The general lesson, stated to a Ministry directly, not softened: a pooled statistic across a stratified population is not evidence about the stratum-level relationship; it can be almost entirely evidence about the stratification itself, and the within-tier residual left after removing that effect can be too small, at this sample size, to read at all. The stratification check is mechanical (runs on every covariate it is pointed at, currently three, always emits a caveat, ADR 0012), so this finding does not depend on remembering to check the one figure that matters this quarter. The composite-index instability is a caveat this document adds on top, since `checks.py` tests declared covariates, not constructed indices, worth naming for Deliverable 3.

Data note, because it matters every time these numbers appear: outcomes in this dataset are synthetic, generated as a function of the capability variables themselves (ADR 0008's honest-limits note), and facility-level mortality here (roughly 14 to 73 per 1,000, pooled per facility across the year) runs three to four times real Rwanda rates. These figures establish what this synthetic sample contains, not what actually drives neonatal mortality in Rwanda, and nothing here should be read as a claim about real Rwandan outcomes.

Deliverable 2, Rapid Prototyping

Working prototype for Problem A: automating the MoH Quarterly Health Bulletin.

Run it

```
cd pipeline
uv sync                           # duckdb, pandas, pyarrow, sf-hamilton
uv run python run.py              # build the mart
cd web && bun install
bun run publish                   # build + publish all four quarters, 2024-Q1 through 2024-Q4
bun run verify                    # build, inline email, run all five checks
open ../output/bulletin-2024-Q1.html
```

Two commands, no server, no API key, no cloud account. Output is one self-contained HTML file with no external assets and no JavaScript.

Solution design (D2 1)

`../artifacts/06-bulletin-architecture-data-flow.html`. Two diagrams: the data-flow pipeline (sources through Bronze/Silver/Gold to the published surfaces), which Sand products are used and why, what is built custom and why, build vs. buy per component; and the data-mart ERD (not a normalized schema, six source files unpivoted through one crosswalk into one canonical fact table).

The written counterpart, same reasoning in prose and tables, standing alone without the HTML artifact, is `SOLUTION-DESIGN.md`.

What it produces

`output/bulletin-<quarter>.html`, one edition per quarter, all four built by `bun run publish`, with every figure carrying its value, its state, and a link to the rows and rules behind it. Lineage records live in the same file as anchors, so it works offline and survives being forwarded.

Four required metrics, two answered directly and two substituted, with the substitution stated rather than papered over:

BRIEF REQUIREMENT	STATUS
Top 10 facilities by volume	rendered, labelled as volume, not quality
Maternal health indicators	**substituted.** The brief names ANC visits, deliveries and complications. The provided CSVs carry deliveries, but no ANC column and no complication column. What ships is neonatal cause of death (ICD-10 P21, P07, P36, Q00 to Q99) and the stillbirth ratio, which are perinatal outcomes, not maternal ones.
Facility performance scores	**partly substituted.** Reporting completeness ships. Timeliness does not and cannot: no source file carries a submission timestamp, so no lag is derivable. What ships instead is governance, operations and staffing capability per tier, section 3. Nothing is scored per facility.
Trend vs previous quarters	shown, caveated: all four quarters, not a two-point delta

What would close the gap. ANC visits, maternal complications and reporting timeliness are all standard DHIS2 data elements. Ingesting them is a Week 2 crosswalk addition, not a pipeline change: the crosswalk already resolves six heterogeneous sources to one grain, and these are three more rows in it. The reason they are absent is the provided extract, not the design. Claiming them as delivered would have been the easier sentence to write and the first thing a reviewer checked.

Facility performance and the trend were the interesting ones to build. `governance.csv` / `operations.csv` / `healthcare_workers.csv` were already resolved through the crosswalk and never queried past three covariates; the trend was withheld because a guard blocked the whole section rather than stating what it could and could not support. Both are gold-layer queries against data already in the mart, not new ingestion. Every check still computes exactly what it always computed; it now always renders its finding, caveated, never gated.

Layout

```
pipeline/
  run.py                  build the mart          Hamilton -> DuckDB -> Parquet
  eda.py                  marimo notebook, all five source files, reuses gold/checks
  web/                    render every surface    Parquet -> HTML
  dataflow/
    bronze.py             source data, verbatim, nothing dropped
    silver.py             canonical observations at DHIS2 grain
    gold.py               the marts a bulletin reads, incl. facility_capability
    checks.py             annotate analyses the data may not fully support
  mart/
    crosswalk.csv               source field -> canonical element      (data, not code)
    org_unit_map.csv            (source, key) -> one org_unit identity (declared, not inferred)
    cause_capability_links.csv  cause of death -> the capability that treats it (declared)
    known_contradictions.csv    crosswalk notes worth stating as findings (declared)
    dhis2_sample.csv            second source, proves the crosswalk
    *.parquet                   published artifacts
output/
  bulletin-*.html         the deliverable, one per quarter
```

Where to put things

ADDING	GOES IN
A new source system	rows in `crosswalk.csv` + a loader in `bronze.py`
A new measure from an existing source	one row in `crosswalk.csv`
A facility identity from another system	one row in `org_unit_map.csv`
A data quality check	`checks.py` , or a `@check_output` on the silver node
A cause-of-death to capability pairing	one row in `cause_capability_links.csv`
A new bulletin panel	a section in `web/src/pages/index.astro`

Three decisions worth reading the code for

`batch` is in the silver key, `dataflow/silver.py` . The sample ships 2024-01 and 2024-03 twice with differing values. Without batch in the key they collide and one silently wins, which is the defect rather than a fix for it. Both are retained; figures drawing on them are provisional and name the conflict.

Identity resolves through a declared table, `mart/org_unit_map.csv`. Not exact-key auto-matching. Two systems using the string `NYA001` is evidence they share a code space, not proof, and fusing on string equality is how an identity graph merges two distinct real entities. An unresolved key raises rather than quietly inventing a facility.

Checks annotate claims, never gate rows, `dataflow/checks.py`. Ordinary validation rejects a bad row. These annotate a *claim*: a trend line on a series with no measured temporal signal, or a pooled correlation that vanishes inside every stratum. The statistics are unchanged from when they gated content (ADR 0008); what changed (ADR 0012) is the consumer contract, a check's finding now always renders with a caveat, never hides the section it qualifies.

Shortcuts taken

- Lineage records are anchors in one file rather than addressable URLs. The contract is the same; the explore surface would serve them at `/metric/.../lineage`.
- The batch collision uses a declared arbitrary default (`DEFAULT-BATCH-01`, lowest occurrence ordinal) so the pipeline completes. Every figure it touches is provisional. Resolution belongs to an analyst, and the queue is specified but not built.
- Gold regenerates fully on every run. Correct and trivially reproducible; would need incremental logic at real scale.
- The stratification check tests against tier only. It should take the stratification variable as a declared input per association.

Further shortcuts (the second source is synthetic, no scheduler, no auth, GPS excluded as fabricated, single-machine DuckDB), the Week 3 plan, and what building this surfaced about the capability-mortality correlation, are in `SOLUTION-DESIGN.md` 3.

Specification

Behaviour is specified in `../openspec/specs/` and `../openspec/changes/`. The specs are deliberately ahead of the code, what is specified and unbuilt is the honest answer to Deliverable 3, rather than five paragraphs of intent.

Checks

`bun run verify` runs five. Four caught a real defect in this build; one is preventive and is marked as such, because a table claiming otherwise would be the first thing to check:

CHECK	HOLDS	CAUGHT
`check-shipped.mjs`	every declared input the pipeline reads by name is tracked by git	`known_contradictions.csv` and `cause_capability_links.csv`, both read by `gold.py` and neither ever added, so a clone raised `FileNotFoundError` on the first command. Every other check ran against the working tree, where they exist
`check-tokens.mjs`	chart theme colours identical to `tokens.css`	nothing yet. Preventive: added when the chart engine moved to a `THEME_SPEC` override, and it guards the same drift class the previous hand-typed palette had
`check-style.mjs`	em dashes, side stripes, script tags, external assets	a latent em dash, and a false positive in its own first rule
`check-email.mjs`	102 KB clip ceiling, no SVG, no flex or grid, state as words	`display:grid` shipped in the email stylesheet
`check-agreement.mjs`	seven shared figures identical across both surfaces	email published 6 withheld panels where the bulletin said 2; later, the trend/correlation caveat wording after the withhold gate was removed

`bun run publish` additionally refuses to write a file whose name and contents disagree on the quarter. That guard exists because the unchecked version shipped `bulletin-2024-Q3.html` containing Q1 figures.

Deliverable 3, Production Hardening

Top five things to fix before this runs unattended for a Ministry.

Each is ranked by what happens if it is skipped, not by effort. Every one is grounded in something this build actually did, and the numbers are reproducible from the repo.

---

1. The batch conflict has no resolution path, and it is silently arbitrary

What happens now. 2024-01 and 2024-03 each arrived twice with differing values, 234 rows, no timestamp, no submission id, no revision flag. Both loads are internally consistent, and the directional bias is 113 rows favouring the first against 119 the second, so nothing in the file establishes which is correct. The pipeline applies `DEFAULT-BATCH-01`, takes the lowest occurrence ordinal, and marks every derived figure provisional. All 30 Q1 district figures carry that mark.

Why it cannot ship as-is. The default is a coin flip wearing a rule name. A quarter where 100% of figures are provisional is honest but not useful, and the Ministry has no way to make it un-provisional.

Fix. The conflicts table and triage queue specified in `data-validation`. A named analyst chooses a batch, the choice is recorded with who and when and why, figures promote from provisional to settled, and the lineage record shows the human decision. This is the difference between a pipeline that discloses a problem and one that lets someone fix it.

Do not resolve this by making the pipeline wait for approval. A blocked pipeline in a Ministry with near-zero IT capacity produces no bulletin at all. Decisions are data, not control flow: the run always completes, and unresolved figures ship provisional.

---

2. Ingestion assumes one file shape, and DHIS2 will not stay still

What happens now. `bronze.py` reads five CSVs whose columns were learned by hand. The crosswalk maps 71 fields across two source systems (66 from the assignment CSVs, 5 from DHIS2). A real deployment ingests DHIS2 exports, HealthTrack dumps, OpenMRS extracts and paper-entry forms, all naming the same field differently, and the mapping table is currently a CSV that a human edits.

Fix. The LLM-proposed, human-confirmed, then cached mapping already argued for in the stack decision. The model proposes a column mapping once, a person confirms it, and it becomes a deterministic entry in the crosswalk. Never re-inferred per run, or the same file parses differently on Tuesday.

Free win first. HDX files carry HXL tags, a machine-readable semantic row above the header. Where the source is HXL-tagged the mapping is a lookup, not a guess, and no model call is needed at all.

---

3. The chart runtime question was answered for the static bulletin, and left open for the surface that doesn't exist yet

What was decided. Flint (`microsoft/flint-chart`) was adopted. `web/src/lib/charts.ts` now authors a Flint spec, compiles it to Vega-Lite (`assembleVegaLite`), and renders it to static SVG with `vega`, headless, in Node at build time. It replaced Observable Plot outright; nothing in this build still calls Observable Plot.

On what grounds. This item originally argued Flint was rejected because it is npm-only with no PyPI package, and the renderer was Python. That objection evaporated on its own schedule: ADR 0010 moved rendering to Astro before this item was ever revisited, so Node and a TypeScript build step were already project dependencies by the time the question came up again. What tipped the decision from "could" to "should" was narrower than the original case for Flint: its theme system (a `swiss` preset, narrowly overridden to this project's exact tokens) removes the literal-hex-duplication problem `check-tokens.mjs` exists to guard against, which a hand-rolled Observable Plot theme could not.

What it actually cost. Flint's semantic layer deliberately does not expose colour/font/tick derivation, so two presentation details had no Flint knob: the hatch fill on provisional bars and the district benchmark rule line. Both are handled by the documented escape hatch, author the Flint spec, then hand-edit the compiled Vega-Lite spec for exactly those two things, not by fighting the grammar. Flint's Vega-Lite compiler also has a real bug: a bar chart using Flint's own auto-generated value-label layers unions the `y` domain across layers and silently falls back to alphabetical sorting under specific multi-layer conditions. Worked around by disabling Flint's auto-labels and hand-authoring one label layer instead; not filed upstream, and worth a minimal repro if it recurs on a chart shape this bulletin does not yet use. Net cost: one theme override that still duplicates hex values against `tokens.css` (same guard, different source shape), one documented compiler workaround, zero new runtime dependencies beyond what ADR 0010 had already added.

What is still open, checked against the code as it stands. The original version of this item flagged that Mosaic (coordinating an explore surface, built on Observable Plot) composing against Flint's Vega-Lite output would mean running two chart runtimes. As of this build, that conflict has not materialised, because the explore surface was never built: it is referenced in the architecture diagram's product-decision cards and in ADR 0012's honest-limits section as "scoped, not implemented." There is exactly one chart runtime in this repo today. That is not the same as the question being resolved, it is the question not having come up yet. ADR 0012 keeps Observable Plot's git history at its parent commit as the named fallback if Flint or Vega prove unable to render server-side outside this session's build environment, which is itself still untested.

Do not build the explore surface against whichever runtime is closest to hand when that work starts. Decide explicitly, in writing, whether it extends Flint/Vega-Lite or brings back Mosaic/Observable Plot for a second runtime, before the first line of that surface is written. A silent default here is exactly the mistake this item was originally written down to avoid, and writing the Flint decision down instead of silently deferring it is the reason it got revisited and actually made.

---

4. Nothing runs on a schedule, and no one is told when it fails

What happens now. `uv run python run.py && bun run publish`, by hand, by someone who knows the command.

Fix. Hamilton already exposes the DAG, so orchestration is a wrapper rather than a rewrite. What matters more than the scheduler choice is the failure surface: a quarterly job that fails silently is worse than no job, because the last good bulletin stays up and looks current. Publication must be gated on the checks passing, and a failed run must reach a person.

The five checks in `web/scripts/` are already the gate. Wire them to the schedule.

---

5. Facility geography is unusable, and the Health Atlas story depends on it

What happens now. `gps_lat` and `gps_lon` are uniform random inside Rwanda's bounding box for all 117 facilities: every province spans the full country extent, and the district column is the only real geography in the file. The bulletin therefore maps at district grain and says so.

Fix. The HDX Rwanda Healthsites layer carries 1,345 real facilities with real coordinates, free and openly licensed. Joining the register to it converts a stated defect into a working facility layer, and it is the precondition for anything the Health Atlas would do, including the browser-local geoprocessing path.

Do not reverse-geocode the district names to synthesise coordinates. That manufactures false precision: a district centroid looks surveyed and cannot be distinguished downstream from a real clinic location. Keep the honest district grain until real points exist.

What is already hardened

Not a to-do list, but the reviewer should know where the line is.

CONCERN	STATE
Analyses the data cannot support	Disclosed with a caveat by two structural checks (`temporal_signal_check` , `stratification_check`), not by reviewer discretion. The statistical bar is unchanged (permutation test, stratified-correlation test, fixed seed); ADR 0012 removed the withhold gate that used to render nothing when a claim failed it, and replaced it with a caveat sentence that always renders. Disclosure, not silence, is now the failure mode.
Figures disagreeing across surfaces	`check-agreement.mjs` , seven shared figures compared in rendered output
A file whose name and contents disagree	`publish.mjs` refuses; the check exists because it happened
Email clipping and unsupported SVG	`check-email.mjs` , measured at 6% of the ceiling
House style violations	`check-style.mjs` , five bans, proven to fail
Chart palette drifting from the stylesheet	`check-tokens.mjs` , nine colours compared

CONCERN	STATE
Provenance of every published figure	Lineage record per figure, section 6
Known defects in the source	Published in the bulletin, section 5, not corrected silently

Deliverable 4, Handover

Reversing ADR 0006: the deliverable is not the artifact, it is a named Digital Health Officer personally restarting the pipeline once, unassisted, before exit. That is the line between real capacity transfer and "shifting the burden": a finished tool handed to a Ministry with near-zero IT capacity degrades post-exit and leaves trust lower than before.

Contents

The three documents below are the answer to "what would you document for the MoH IT team". They are written, not proposed. They ship in the repository at `deliverable-4-handover/` rather than in this PDF, because at nine pages they would have made Parts 3 and 4 a third of a document whose brief asks for 80% of the effort on Parts 1 and 2.

- `runbook.md` : the four-command quarterly cycle, what each check failing means, how to recover from a failed run.
- `data-contract.md` : the required input shape, the crosswalk, and precisely what happens when a source column is renamed, absent, or new (verified by running each case, not inferred).
- `decision-register.md` : what happens to a batch conflict today (`DEFAULT-BATCH-01` , unconditional, provisional), and what is specified but not built (the triage queue).

Exit criterion

A named DHO, given only these three documents and no live assistance, runs the full quarterly cycle from a clean checkout and produces a bulletin that passes all five checks.

Not yet satisfied. This requires a named person with cleared hours, which ADR 0006's own kill criteria treat as an unconfirmed empirical fact rather than an assumption: "Nobody has asked whether a named DHO has cleared hours." What this deliverable can and does verify is the half of the criterion within this repo's control: that the runbook itself is complete enough for an unassisted restart to succeed, by following it literally against a fully wiped state. See "Verification" below.

Training plan

Two sessions, because the runbook is four commands and the judgment calls are what actually need a person, not a memorised sequence.

Session 1 (60 to 90 min): run the cycle. Screen-share, the DHO drives, I do not touch the keyboard. `runbook.md` end to end against last quarter's real data: sync, rebuild, publish, verify. The DHO reads every line the terminal prints, including a full checks pass, so "verify passed" has a face they have seen, not a green checkmark they trust blind.

Session 2 (45 to 60 min): read a failure. I break something in a scratch copy in front of them (a renamed CSV column, per `data-contract.md` ; a check made to fail on purpose, then restored) and the DHO diagnoses it using only `runbook.md` 's recovery table. Success is the DHO stating which document to open before I tell them, not fixing the specific break, since next quarter's break will not be this one.

Between sessions: the DHO gets read access to this repo, `data-contract.md`, and one week to sit with the material before session 2, which is deliberately the same cadence ADR 0006's Week 1 Day 2 action treats as the earliest honest checkpoint on this handover.

Verification

Run 2026-08-11, from a fully wiped state (`pipeline/mart`, `web/dist`, `web/node_modules`, `pipeline/.venv` all removed), following `runbook.md`'s "Full clean rebuild" section with no step added, removed, or reordered from what is written there:

```
uv sync                                  ok
uv run python run.py                     ok, 10 parquet files written
bun install                              ok
bun run publish                          ok, both quarters published
bun run verify                           5/5 checks passed
```

`output/bulletin-2024-Q1.html` regenerated at the same byte hash (`05ce82f7889cc85f38ec61cdf3d4e0ab`) as the previously committed copy, on a rerun with no inputs changed, confirming the pipeline is deterministic rather than merely "usually working."

This run predates ADR 0012 and is stale. The withhold-gate removal, the four-quarter build, the four new gold-layer tables, and the Flint chart rewrite all landed after 2026-08-11 and change what this command sequence produces: `run.py` now writes 14 parquet files (`facility_capability`, `capability_summary`, `cause_capability_links` and `known_contradictions` joined the original ten), and `bun run publish` builds all four quarters, not two. The committed `output/bulletin-2024-Q1.html` at the current commit hashes to `2bafa326c8925bfe508bc337e97e9e6e`, which does not and should not match the 2026-08-11 hash above; the underlying document changed on purpose. What this run does **not** re-establish is the determinism claim itself: I did not rerun the full clean rebuild against the post-ADR-0012 code for this pass, specifically to avoid writing into `deliverable-2-prototype/` while it has other work in flight. Re-running `runbook.md`'s "Full clean rebuild" section once, twice, and diffing the two hashes is the next action before this deliverable can claim reproducibility again; until then, treat determinism as carried over by design but **not measured** post-rework.

One correction to the sentence above, since it was measured after this was first written and the earlier wording would not survive a reviewer running the pipeline twice. The mart is **not** byte-identical between runs, and by design: `bronze.py` stamps every row with `ingested_at = datetime.now(timezone.utc)`, one timestamp per run, so that one run is one provenance event. Two consecutive runs therefore produce differing `silver.parquet` and `observations_resolved.parquet`. Excluding that column and sorting, every other column is byte-identical. So the accurate claim is that the pipeline is deterministic **in its outputs given its inputs, excluding the provenance timestamp it deliberately varies**, not that the files hash the same. The rendered bulletin is unaffected, because no figure reads `ingested_at`. This proves the runbook's commands were sufficient for a rebuild from nothing as of 2026-08-11. It never proved, and still cannot prove, that a specific named person can execute them unassisted; that half of the exit criterion is the confirmation named in "Exit criterion" above, still open.

What is deliberately not here

No architecture diagram beyond what `decisions/` and `openspec/specs/` already carry; duplicating it here would be a second copy that drifts from the first. No slide deck: the brief asks for a runbook the DHO can act on at 2am with no one to call, and a deck is not that document.

NEONATAL MORTALITY, 2024-Q1

50.6 per 1,000 live births

50.6 per 1,000 is 2.7× the WHO / national benchmark of 19 per 1,000.

1,819 deaths across **35,916 live births**, WHO GHO definition, 2 of 3 expected months, 2024-02 absent. 30 of 30 district figures are PROVISIONAL: 2024-01 and 2024-03 each loaded twice with differing values. Only 1 of 117 facilities stock surfactant, the standard treatment for prematurity; only 9 of 117 run an active quality-improvement programme; median self-reported reporting completeness is 76%. Capability gaps qualify whether this figure is actionable, not only whether it is settled.

Resolve the 2024-01 and 2024-03 batch conflict before treating this figure as final: DEFAULT-BATCH-01 is a placeholder default, not a decision.

MINISTRY OF HEALTH, RWANDA · NEONATAL · 2024-Q1

Quarterly Health Bulletin

2024-Q1

MONTHS HELD

2/3

2024-02 absent

FACILITIES REPORTING

117/117

in the months held

FIGURES PROVISIONAL

30/30

awaiting a batch decision

REPORTING COMPLETENESS, MEDIAN

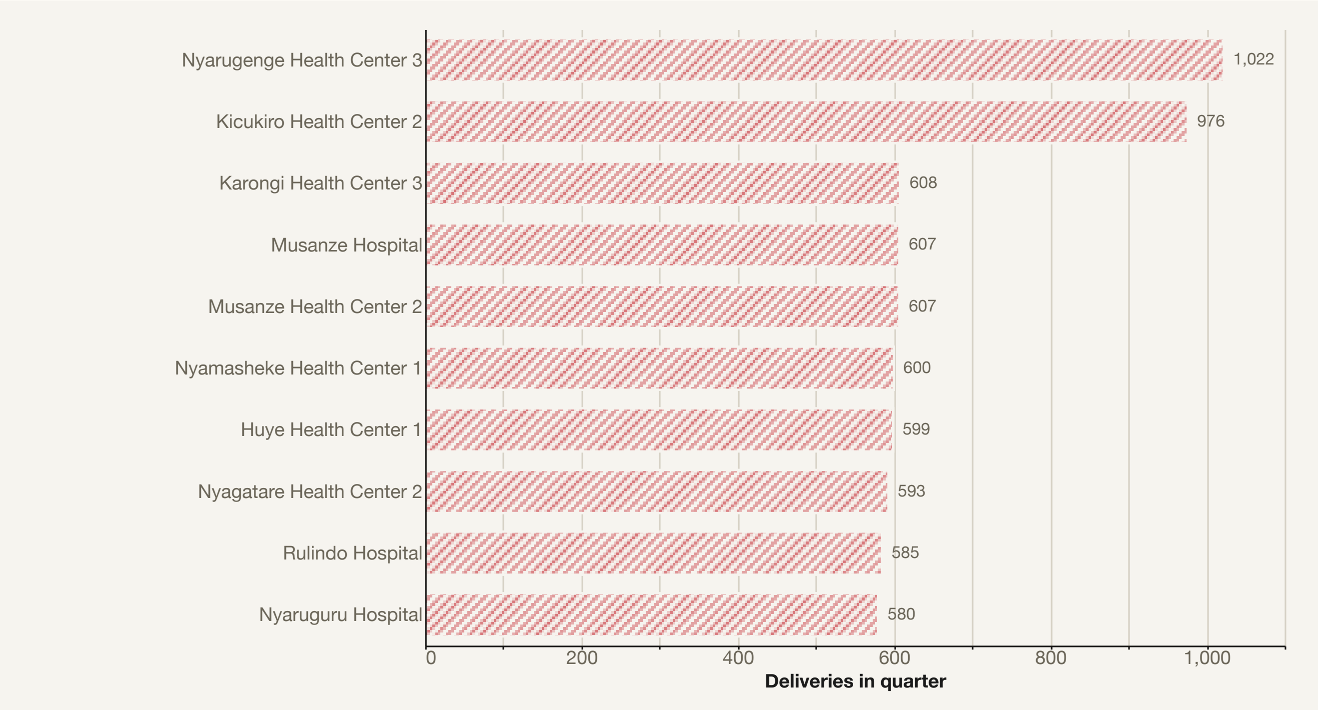
76%

range 50% to 99%, all facilities

Provisional figures are hatched in bars, hollow in dots, underlined in tables. Every check below always renders its finding: a caveat qualifies confidence, it never withholds content. Every figure links to its lineage record.

01

Facilities by delivery volume

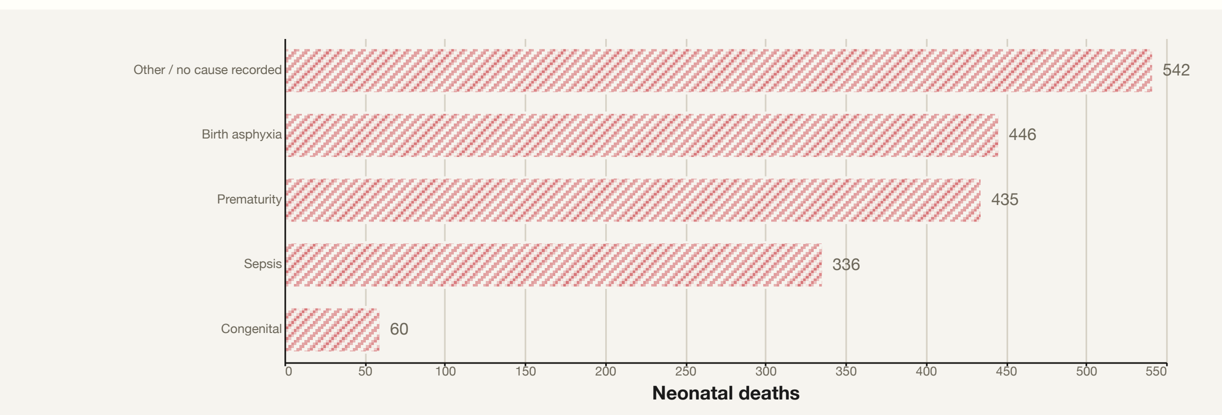


Top 10 of 117 by total deliveries. Volume tracks tier, so this is not a quality ranking. lineage, section 8

01

Neonatal deaths by cause, paired with the capability that treats it

The largest single bucket is **Other / no cause recorded**, 542 of 1,819 deaths (29.8%), ahead of every named clinical cause below it. That is a data-system finding, not a clinical one: it means the cause of death was not recorded, not that no cause exists. A reader taking the largest named clinical cause, Birth asphyxia, as the leading cause of neonatal death in this data would be wrong.



Ranked by count, not by assumed clinical severity. lineage, section 8

CAUSE	ICD-10	DEATHS	SHARE
Other / no cause recorded	P00–P96	542	29.8%

CAUSE	ICD-10	DEATHS	SHARE
Birth asphyxia Median competency 36% (range 10% to 96%) across 117 facilities	P21	446	24.5%
Prematurity Available at 1 of 117 facilities	P07	435	23.9%
Sepsis Usually or Always available at 14 of 117 facilities	P36	336	18.5%
Congenital Available at 10 of 117 facilities	Q00–Q99	60	3.3%

Reconciles to total neonatal deaths in 1,404 of 1,404 source rows. Capability figures are national, not quarter-scoped: capability data is a facility snapshot, section 3. lineage, section 8

Stillbirths, 2024-Q1: a different denominator, not a sixth cause row above. A stillbirth never had a live birth, so it cannot enter a live-birth-denominated rate and is not added to the neonatal-death count.

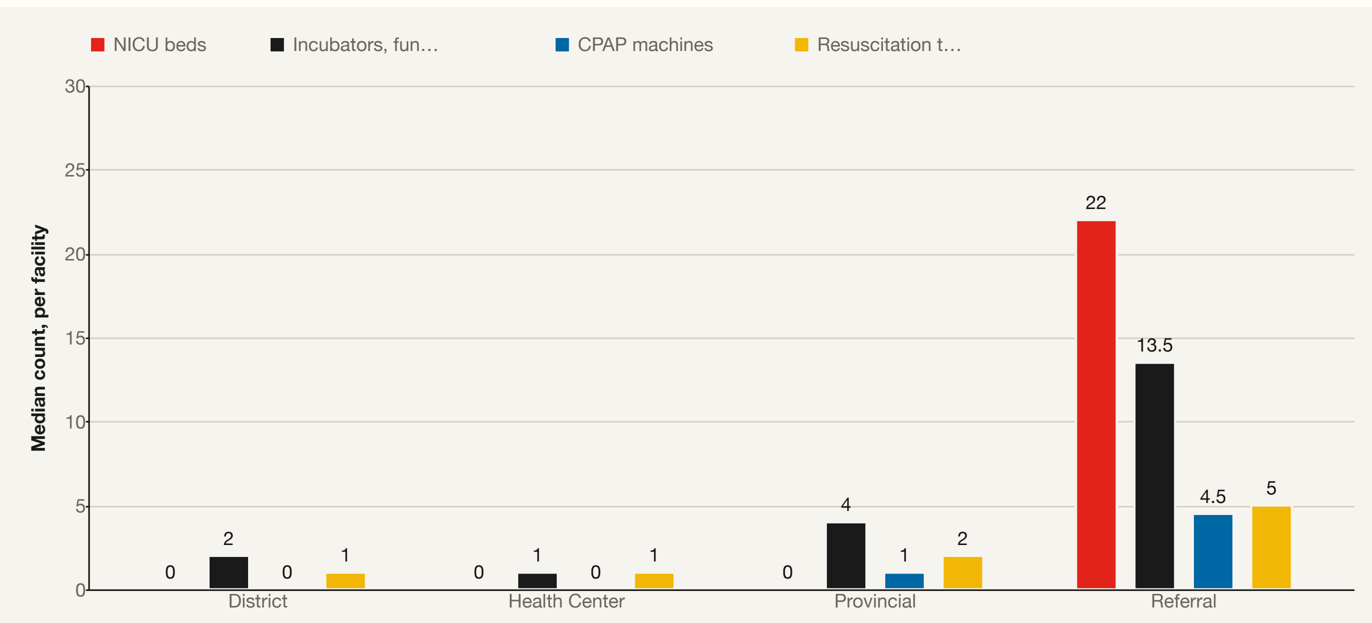
27.3

per 1,000 total deliveries · 1,008 of 36,924 · provisional

03

Facility capability, national, n = 117

Governance, operations and staffing data from all 117 facilities, resolved through the same crosswalk as every other figure on this page but never previously queried past the three covariates section 6 tests. Reporting completeness (median 76%) is the brief's own required "facility performance score." Broken out by tier below because tier, not any single equipment field, is what actually separates facilities in this data (section 6): District and Health Center read as near-identical on every metric in the table below, despite the widest mortality gap of any two adjacent tiers.



Median count per facility, by tier, NICU beds, Incubators, functional, CPAP machines, Resuscitation tables. lineage, section 8

06

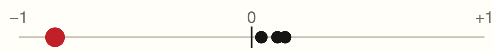
Tier confound: capability looks predictive, until you stratify

District facilities, nominally the mid-tier, have the worst mean neonatal mortality of any tier: 67.4 per 1,000 (n=45), ahead of Health Center (62.2, n=45), Provincial (32.5, n=25) and Referral (14.7, n=2).

A composite equipment index correlates -0.87 with facility mortality pooled across all 117 facilities, a relationship strong enough to read as "equip facilities, save newborns." It is not that. Stratified by tier it collapses to near zero or reverses (District -0.071; Health Center +0.036; Provincial -0.288): District and Health Center hold almost identical equipment (section 3) and still land 5.2 points apart in mortality. The pooled number describes which tier a facility is in, not what equipping it would do. Every covariate below is tested the same way, pooled and within tier: shown here, not hidden. Every correlation below also tests the same static, national capability snapshot (section 3) against each facility's births-weighted mortality rate summed across all four quarters of 2024, not matched quarter by quarter against that snapshot.

Referral closes a similar gap between two figures that sound alike. Whether a referral gets a reply pools at -0.81 against mortality; how fast it happens pools at +0.13. Neither survives stratification (below): the loop closing and the loop's speed both read as tier artifacts in this data, not real drivers of mortality once tier is held constant.

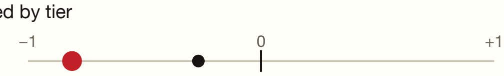
Staff trained on protocol (%) vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	−0.844
District	+0.111
Health Center	+0.042
Provincial	+0.144

Caveat: pooled r=-0.844 does not survive stratification by tier (District +0.111; Health Center +0.042; Provincial +0.144); the association describes tier, not the covariate

CPAP machines vs. neonatal mortality rate, stratified by tier



STRATUM	R
pooled, all facilities	−0.811
Provincial	−0.269

Caveat: pooled r=-0.811 does not survive stratification by tier (Provincial -0.269); the association describes tier, not the covariate